

Notes: House and Shapiro (AER, 2008)
**Temporary Investment Tax Incentives: Theory with Evidence from Bonus
Depreciation**

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1 Big picture

- **Question.** How much can a temporary investment tax incentive stimulate investment, and what does the response reveal about the supply elasticity of investment goods?
- **Policy setting.** The paper studies the 2002 and 2003 US bonus depreciation provisions. The 2002 law allowed firms to deduct 30 percent of eligible investment immediately, while the 2003 law raised the immediate deduction to 50 percent. The policy was explicitly temporary and applied only to capital with MACRS recovery periods of 20 years or less.
- **Core theoretical insight.** For long-lived capital goods, the intertemporal elasticity of investment is nearly infinite. Firms can move future purchases into the subsidy window, so a temporary investment subsidy should be fully capitalized into the shadow price of investment goods.
- **Empirical strategy.** The policy created cross-sectional variation across capital types because the subsidy was more valuable for long-lived qualifying capital, and zero for nonqualifying structures. The authors compare investment and price forecast errors across capital types and policy periods.
- **Main quantity result.** Investment in the capital categories that benefited most from bonus depreciation rose sharply relative to less subsidized or nonqualified capital.
- **Main structural result.** The estimated investment supply elasticity is high, generally in the range of about 6 to 14 across specifications.
- **Main price result.** The data do not show a clear market price response to the subsidy. The paper interprets this as consistent with either internal adjustment costs or measurement error in capital-goods price data that is large relative to the expected price changes.
- **Economic takeaway.** Temporary tax incentives can strongly shift the timing and composition of investment. Quantity responses contain useful information about investment supply because theory pins down the shadow-price response.

2 Incremental contribution of the paper

- **From permanent cost-of-capital analysis to temporary policy analysis.** Standard user-cost reasoning often focuses on permanent tax changes. This paper emphasizes that temporary incentives are different because firms can intertemporally substitute investment into the subsidy window.
- **A sharp theoretical restriction.** The paper shows that, for sufficiently long-lived capital and sufficiently temporary subsidies, the shadow price of investment rises approximately one-for-one with the subsidy regardless of the supply elasticity.
- **A new interpretation of price pass-through.** Observing higher investment-good prices after a temporary subsidy is not evidence of inelastic supply. Under the model, shadow prices should move with the subsidy even if supply is elastic.
- **Identification from quantities alone.** Because the model pins down the shadow-price response, the supply elasticity can be estimated from investment quantity responses rather than relying on noisy market-price changes.

- **Use of a natural experiment.** Bonus depreciation created differential treatment by recovery period and eligibility status, producing a policy shock that maps cleanly into the theory.
- **Evidence on adjustment costs.** The contrast between strong quantity responses and weak observed price responses sheds light on whether adjustment costs are external market-price costs or internal costs borne inside firms.
- **Policy relevance.** The paper provides quantitative evidence on a widely used countercyclical fiscal tool and clarifies when temporary tax incentives are likely to have large short-run effects.

3 Institutional background and measurement

3.1 Bonus depreciation

- The Job Creation and Worker Assistance Act of 2002 introduced a 30 percent bonus depreciation allowance.
- The Jobs and Growth Tax Relief Reconciliation Act of 2003 increased the bonus allowance to 50 percent.
- Under the policy, firms could deduct the bonus share immediately and then depreciate the remaining basis under the normal MACRS schedule.
- Eligible property had to be depreciable under MACRS and have a recovery period of 20 years or less.
- The policy applied to property placed in service after September 11, 2001 and before January 1, 2005, with some timing extensions for certain long-lived projects.

Interpretation: The policy was temporary and uneven across capital types. That is exactly the kind of environment in which the authors’ intertemporal substitution logic should be powerful.

3.2 Tax lives and treatment intensity

- Capital types differ in MACRS recovery periods and depreciation methods.
- Short-lived equipment such as computers and vehicles already receives relatively accelerated depreciation, so the incremental value of bonus depreciation is small.
- Long-lived qualifying capital, such as 15-year and 20-year property, receives a larger incremental benefit.
- Nonresidential real property has a 39-year recovery period and does not qualify, providing a useful comparison group of long-lived but unsubsidized capital.

Interpretation: The strongest treatment is for “quasi-structures” - long-lived assets that qualify for bonus depreciation. This makes the empirical design closer to a dose-response design than a simple treated-versus-control comparison.

3.3 Investment and price data

- The authors use type-specific data on real investment quantities and real relative prices.
- They form forecast errors using pre-policy forecasting equations and then relate these errors to the tax depreciation rate, the bonus rate, and eligibility.
- The quantity data show systematic movement in the predicted direction.
- Price data are much noisier; the predicted maximum price effect is only a few percentage points for even the most heavily subsidized capital goods.

Interpretation: The paper’s empirical strength comes from the cross-section of capital types. The price tests are conceptually important but empirically underpowered because capital price indexes are noisy.

4 Model and economic mechanism

4.1 Firm problem

The firm chooses investment in different capital types. Let m index capital types, I_t^m denote gross investment, K_t^m the capital stock, δ_m economic depreciation, φ_t^m the pre-tax shadow price of investment goods, and ζ_t^m the effective tax subsidy on new capital.

A simplified version of the investment first-order condition is

$$q_t^m = u'(C_t)\varphi_t^m(1 - \zeta_t^m),$$

where q_t^m is the shadow value of installed capital.

Interpretation: A temporary subsidy lowers the after-tax price of investment. For long-lived capital, firms can arbitrage this temporary price movement by shifting investment timing.

4.2 Investment supply

The paper uses a type-specific investment supply curve of the form

$$\varphi^m(I_t^m) = \left(\frac{I_t^m}{\bar{I}^m} \right)^{1/\xi},$$

where ξ is the supply elasticity.

- A high ξ means a large quantity response for a given shadow-price change.
- A low ξ means a steep supply curve and smaller quantity response.
- The model allows the price wedge to represent external market-price costs, internal adjustment costs, or a combination.

Interpretation: The empirical question is not whether firms respond to the subsidy - the model predicts they should - but how large the quantity response is for the model-implied change in the shadow price.

4.3 Key approximation for temporary policies

For long-lived capital and short-lived tax incentives, the installed-capital shadow value is approximately unchanged:

$$q_t^m \approx \bar{q}^m.$$

Thus the pre-tax shadow price must move approximately one-for-one with the tax subsidy:

$$\varphi_t^m \approx \frac{\bar{q}^m}{u'(C_t)(1 - \zeta_t^m)}.$$

Taking logs and linearizing gives an investment response like

$$\hat{I}_t^m = \frac{\xi}{1 - \zeta^m} d\zeta_t^m + \frac{\xi}{\sigma} \hat{C}_t,$$

where the second term captures aggregate marginal-utility effects.

Interpretation: The model implies that cross-capital variation in the value of the subsidy should map directly into cross-capital variation in investment responses.

4.4 Bonus depreciation in the model

Let z^m be the present value of normal depreciation deductions and λ_t^m the bonus depreciation rate. The effective change in the subsidy is proportional to

$$(1 - z^m)\lambda_t^m.$$

- The bonus is more valuable when normal depreciation deductions are slow.
- Hence the subsidy is larger for 15-year and 20-year qualifying capital than for 3-year or 5-year capital.
- Nonqualified 39-year structures receive no bonus, despite being long-lived.

Interpretation: Bonus depreciation generates a clean ranking: long-lived qualifying assets should respond most, short-lived assets should respond less, and nonqualified structures should not respond mechanically to the bonus.

5 Empirical specifications

5.1 Forecast-error construction

The empirical analysis first estimates baseline forecasting equations for real investment and real relative prices. The authors then compute forecast errors by capital type and period.

Conceptually,

$$\text{forecast error}_{m,t} = \text{actual}_{m,t} - \widehat{\text{normal}}_{m,t}.$$

- The forecast errors are compared across five periods: pre-policy, anticipation, 30 percent bonus, 50 percent bonus, and post-policy.

- The tax depreciation rate summarizes the relevant tax life of each capital type.
- Qualified capital is marked separately from nonqualified capital.

Interpretation: The empirical design asks whether investment forecast errors line up with the predicted subsidy intensity across capital types and policy windows.

5.2 Structural investment equation

The investment equation relates real investment forecast errors to the value of bonus depreciation. The key coefficient estimates ξ , the investment supply elasticity.

- The baseline WLS estimate of ξ is 7.68.
- Alternative specifications give estimates roughly between 6 and 14.
- The estimates are generally precise enough to reject small elasticities.

Interpretation: Investment supply appears highly elastic in the relevant short-run policy experiment.

5.3 Structural price equation

The price equation asks whether observed real relative prices move with the subsidy. If all adjustment costs are external and appear in observed purchase prices, the price coefficient should be close to one. If adjustment costs are internal, the market-price coefficient can be close to zero.

- The price coefficients are generally negative and imprecise.
- The data reject neither full external adjustment costs nor no external adjustment costs.
- The likely reason is that expected price effects are small relative to the noise in type-specific price indexes.

Interpretation: Quantity data provide the cleaner evidence. Price data are informative conceptually but too noisy to distinguish market pass-through from internal adjustment costs.

6 Identification and empirical design

6.1 What identifies the policy effect

- **Time variation:** the 30 percent bonus, the 50 percent bonus, and the expiration period provide policy timing.
- **Cross-sectional variation:** different capital types receive different effective subsidies because of tax recovery periods and eligibility.
- **Dose-response logic:** the largest predicted responses occur for qualified long-lived capital.
- **Nonqualified comparison:** nonresidential structures are long-lived but do not qualify, helping separate long-life effects from subsidy eligibility.

Interpretation: The most persuasive part of the design is the cross-sectional dose response across capital types during the policy windows.

6.2 Key assumptions

- The policy shock is temporary and credible enough to generate intertemporal substitution.
- The approximation $q_t^m \approx \bar{q}^m$ is accurate for the relevant long-lived goods.
- Forecast errors would not systematically line up with tax depreciation rates in the absence of the policy.
- The investment supply function is stable across capital types and policy windows.

Interpretation: The design is not a randomized experiment, but the theory yields strong predictions about exactly which capital types should respond.

6.3 Threats and limitations

- **Anticipation.** Firms may have responded before formal enactment because the policy was expected and was made retroactive.
- **Expiration timing.** Some long-lived projects had long construction lags, and Section 179 expensing complicates the end of the policy window.
- **Measurement error.** Type-specific price data are noisy, especially for structures and quasi-structures.
- **Aggregate effects.** The policy may have modest aggregate effects, although the paper controls for aggregate marginal-utility movements.

Interpretation: These concerns weaken the time-series evidence more than the cross-sectional evidence. The strongest evidence remains the pattern across capital types.

7 Tables and figures

7.1 Figure 1 and Table 1: core theory and approximation

Read:

- Figure 1 shows why the shadow price rises by the tax subsidy. The horizontal after-tax valuation condition shifts up, and investment increases along the supply curve.
- A flatter supply curve, corresponding to a higher ξ , produces a larger quantity response for the same shadow-price movement.
- Table 1 evaluates how accurate the approximation is for different subsidy durations, depreciation rates, and supply elasticities.
- The approximation is very good for temporary policies and low-depreciation capital. It becomes worse when policies last longer, depreciation is high, or the supply elasticity is very high.

Economic message: The central logic is intertemporal arbitrage. For long-lived capital, a temporary subsidy primarily changes when firms buy capital rather than the fundamental value of having capital installed.

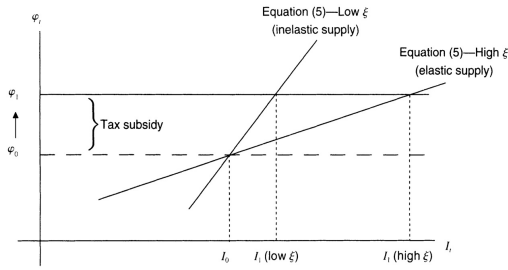


FIGURE 1. PRICE AND QUANTITY RESPONSES TO TEMPORARY INVESTMENT SUBSIDIES

TABLE 1—RESPONSE TO A TEMPORARY INVESTMENT SUBSIDY

Duration	Depreciation rate	Shadow price (φ)						
		$\xi = 0$	$\xi = 0.5$	$\xi = 1$	$\xi = 5$	$\xi = 10$	$\xi = 15$	$\xi = 20$
6 months	$\delta = 0.001$	1.000	1.000	1.000	0.999	0.998	0.997	0.996
	$\delta = 0.01$	1.000	0.999	0.998	0.992	0.986	0.982	0.978
	$\delta = 0.02$	1.000	0.998	0.996	0.986	0.976	0.969	0.963
	$\delta = 0.05$	1.000	0.996	0.992	0.970	0.951	0.936	0.923
	$\delta = 0.10$	1.000	0.992	0.985	0.945	0.911	0.885	0.864
	$\delta = 0.25$	1.000	0.982	0.965	0.877	0.807	0.755	0.714
1 year	$\delta = 0.001$	1.000	1.000	0.999	0.997	0.995	0.993	0.991
	$\delta = 0.01$	1.000	0.998	0.996	0.983	0.972	0.964	0.956
	$\delta = 0.02$	1.000	0.996	0.993	0.972	0.954	0.940	0.928
	$\delta = 0.05$	1.000	0.992	0.984	0.941	0.906	0.878	0.855
	$\delta = 0.10$	1.000	0.985	0.971	0.896	0.835	0.790	0.753
	$\delta = 0.25$	1.000	0.966	0.936	0.784	0.673	0.597	0.539
2 years	$\delta = 0.001$	1.000	0.999	0.999	0.995	0.990	0.986	0.983
	$\delta = 0.01$	1.000	0.996	0.992	0.967	0.946	0.930	0.915
	$\delta = 0.02$	1.000	0.992	0.985	0.946	0.912	0.886	0.864
	$\delta = 0.05$	1.000	0.984	0.969	0.890	0.826	0.779	0.740
	$\delta = 0.10$	1.000	0.971	0.946	0.814	0.715	0.645	0.591
	$\delta = 0.25$	1.000	0.941	0.891	0.659	0.515	0.428	0.368
3 years	$\delta = 0.001$	1.000	0.999	0.998	0.992	0.985	0.980	0.975
	$\delta = 0.01$	1.000	0.993	0.988	0.952	0.922	0.898	0.878
	$\delta = 0.02$	1.000	0.989	0.979	0.921	0.873	0.837	0.807
	$\delta = 0.05$	1.000	0.976	0.956	0.845	0.760	0.698	0.649
	$\delta = 0.10$	1.000	0.959	0.925	0.749	0.626	0.545	0.485
	$\delta = 0.25$	1.000	0.922	0.860	0.587	0.439	0.357	0.304
Permanent	$\delta = 0.001$	1.000	0.986	0.972	0.884	0.806	0.749	0.704
	$\delta = 0.01$	1.000	0.929	0.872	0.637	0.513	0.443	0.396
	$\delta = 0.02$	1.000	0.908	0.839	0.578	0.453	0.387	0.343
	$\delta = 0.05$	1.000	0.888	0.808	0.528	0.405	0.341	0.300
	$\delta = 0.10$	1.000	0.879	0.794	0.506	0.384	0.322	0.282
	$\delta = 0.25$	1.000	0.872	0.783	0.489	0.367	0.306	0.267

Notes: The table shows the equilibrium percent change in the shadow price of capital goods φ in response to an investment subsidy of 1 percent ($d\xi = 0.01$). Investment supply is given by equation (5). For the numerical calculations, the production function is AK^α , $r = 0.02$, and $\alpha = 0.35$.

(a) Figure 1: price and quantity responses to temporary investment subsidies. (b) Table 1: accuracy of the temporary-policy approximation.

7.2 Table 2 and Table 3: tax lives and the size of the bonus subsidy

Read:

- Table 2 maps capital categories to MACRS recovery periods and approximate tax depreciation rates.
- Computers, office equipment, vehicles, buses, and trucks have short recovery periods, while radio towers, pipelines, utilities, railroad structures, and farm buildings have longer qualifying lives.
- Nonresidential real property has a 39-year recovery period and does not qualify for bonus depreciation.
- Table 3 shows that the value of bonus depreciation is increasing in the recovery period and in the nominal interest rate.
- With a 50 percent bonus and a 5 percent nominal rate, the tax subsidy is only 1.26 percent for 5-year property but 4.78 percent for 20-year property.

Economic message: The law created a stronger incentive for long-lived qualifying capital than for short-lived equipment. This differential subsidy is what powers the empirical design.

TABLE 2—RECOVERY PERIODS AND DEPRECIATION METHODS BY TYPE OF CAPITAL

Type of capital	Recovery period, R (years)	Tax depreciation rate, δ (percent)	Method
Tractor units for over-the-road use, horses over 12 years of age or racehorses with over 2 years in service	3	66.7	200 DB
Computers and office equipment; light vehicles, buses and trucks	5	40.0	200 DB
Miscellaneous equipment, office furniture, agricultural equipment	7	28.6 or 21.4	200 DB or 150 DB
Water transportation equipment (vessels and barges); single-purpose agricultural structures	10	20.0 or 15.0	200 DB or 150 DB
Radio towers, cable lines, pipelines, electricity generation and distribution systems, "land improvements," e.g., sidewalks, roads, canals, drainage systems, sewers, docks, bridges, engines and turbines	15	10.0	150 DB
Farm buildings (other than single purpose structures), railroad structures, telephone communications, electric utilities, water utilities structures including dams, and canals	20	7.5	150 DB
Nonresidential real property (office buildings, storehouses, warehouses, etc.)	39	2.6	SL

Note: Tax depreciation methods are 200 percent declining balance (200 DB), 150 percent declining balance (150 DB), and straight line (SL).

Source: IRS Publication 946.

TABLE 3—QUANTIFYING DEPRECIATION ALLOWANCES

Recovery period	Nominal interest rate = 0.03			Nominal interest rate = 0.05			Nominal interest rate = 0.07		
	$\lambda^n = 0$	$\lambda^n = 0.3$	$\lambda^n = 0.5$	$\lambda^n = 0$	$\lambda^n = 0.3$	$\lambda^n = 0.5$	$\lambda^n = 0$	$\lambda^n = 0.3$	$\lambda^n = 0.5$
<i>Panel A: Present value of depreciation allowances: $\lambda^n + (1 - \lambda^n)z^n$</i>									
3 years	0.972	0.981	0.986	0.955	0.968	0.977	0.939	0.957	0.969
5 years	0.949	0.964	0.975	0.918	0.943	0.959	0.890	0.923	0.945
7 years	0.927	0.949	0.964	0.884	0.919	0.942	0.846	0.892	0.923
7 years (150DB)	0.914	0.939	0.957	0.863	0.904	0.932	0.818	0.872	0.909
10 years	0.896	0.927	0.948	0.837	0.886	0.919	0.786	0.850	0.893
10 years (150DB)	0.878	0.915	0.939	0.811	0.868	0.905	0.752	0.826	0.876
15 years	0.824	0.877	0.912	0.733	0.813	0.867	0.659	0.761	0.829
20 years	0.775	0.842	0.887	0.667	0.767	0.833	0.582	0.708	0.791
<i>Panel B: Tax subsidy due to the bonus depreciation allowance, percent</i>									
3 years	0.0	0.26	0.44	0.0	0.42	0.70	0.0	0.57	0.95
5 years	0.0	0.48	0.79	0.0	0.76	1.26	0.0	1.01	1.69
7 years	0.0	0.68	1.13	0.0	1.06	1.77	0.0	1.40	2.33
7 years (150DB)	0.0	0.80	1.33	0.0	1.25	2.08	0.0	1.64	2.73
10 years	0.0	0.96	1.60	0.0	1.47	2.45	0.0	1.91	3.18
10 years (150DB)	0.0	1.11	1.86	0.0	1.70	2.83	0.0	2.19	3.65
15 years	0.0	1.58	2.64	0.0	2.34	3.89	0.0	2.93	4.88
20 years	0.0	2.00	3.33	0.0	2.87	4.78	0.0	3.51	5.85

(a) Table 2: recovery periods and depreciation methods by capital type. (b) Table 3: present value of depreciation allowances and implied subsidy.

7.3 Figure 2 and Figure 3: predicted responses and empirical forecast errors

Read:

- Figure 2 simulates the model response to the 30 percent and 50 percent bonus depreciation policies.
- Investment responses are largest for qualified capital with low tax depreciation rates, especially 15-year and 20-year property.
- The price panels show the same ranking scaled down by the supply elasticity.
- Figure 3 plots actual forecast errors for investment quantities and real prices.
- The investment panels show the predicted cross-sectional pattern: larger positive forecast errors for heavily subsidized capital during the bonus periods.
- The price panels are much noisier and do not show the same clean pattern.

Economic message: The model's predicted quantity pattern appears clearly in the data. The price evidence is weak not because the theory is irrelevant, but because observed price indexes are too noisy relative to the expected price change.

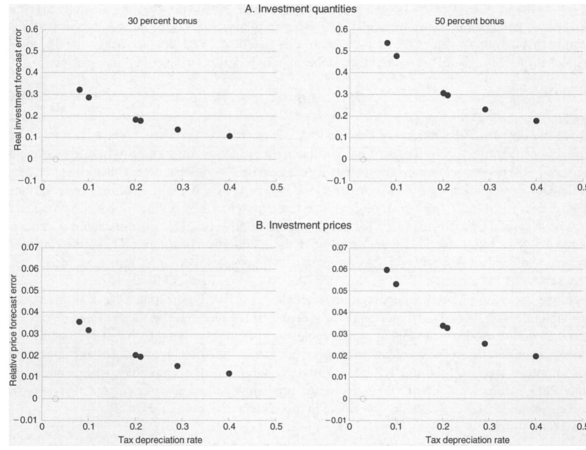


FIGURE 2. SIMULATED RESPONSE TO BONUS DEPRECIATION POLICY

Notes: Simulated response of investment (top panels) and shadow prices (lower panels) for various types of capital to the 30 percent (2002:II) and 50 percent (2003:II) bonus depreciation policy. The approximate geometric tax depreciation rate (δ^*) is on the horizontal axis. Percent deviation from steady state is on the vertical axis. Each circle corresponds to approximate response to bonus depreciation based on equations (17) and (18). Solid circles are for capital that qualifies for bonus depreciation. Empty circles represent unqualified capital. In the upper panels, $\xi = 9$.

(a) Figure 2: simulated response to bonus depreciation.

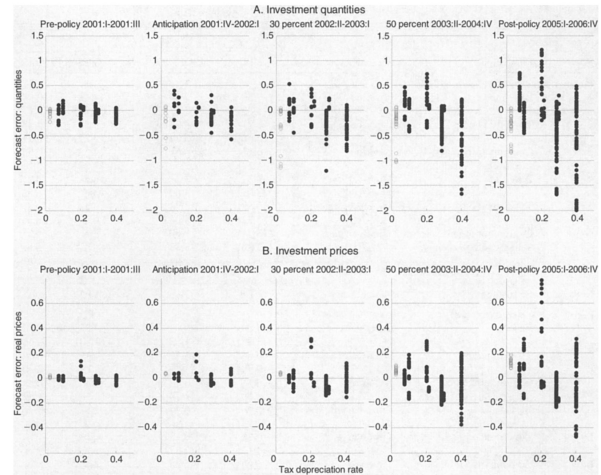


FIGURE 3. FORECAST ERRORS FOR REAL INVESTMENT AND REAL PRICES

Notes: The figure plots forecast errors for real investment (upper panels) and real investment prices (lower panels) by type of capital. The forecast errors come from the baseline forecasting equations (19) and (20). Solid circles are for capital that qualify for bonus depreciation. Empty circles are unqualified capital. The tax depreciation rate (δ^*) is on the horizontal axis.

(b) Figure 3: forecast errors for real investment and real prices.

7.4 Table 4 and Figure 4: structural estimates and time profile

Read:

- Table 4 reports the structural parameter estimates for the investment equation and price equation.
- In the investment equation, the baseline WLS estimate of ξ is 7.68. Other WLS estimates are 6.31, 6.13, 14.82, 13.83, and 13.21.
- These estimates imply a highly elastic short-run supply response to the model-implied subsidy.
- In the price equation, the estimated pass-through coefficients are negative and imprecise.
- Figure 4 estimates the time path of the implied bonus depreciation rate. The estimated bonus rises during the anticipation and policy periods, but the expiration pattern is not as sharp as the cross-sectional investment pattern.

Economic message: The structural estimates confirm the visual evidence. Bonus depreciation increased investment substantially, but observed price data are too noisy to pin down whether adjustment costs are external or internal.

TABLE 4—STRUCTURAL PARAMETER ESTIMATES

			ξ		ξ/σ	
Row	First stage	Second stage	WLS	OLS	WLS	OLS
Panel A: Investment equation (21)						
1	Baseline	Baseline	7.68 (1.85)	7.03 (2.67)	14.41 (2.79)	10.04 (5.77)
2	Time trend only	Baseline	6.31 (3.66)	7.09 (5.87)	13.89 (5.27)	14.58 (8.73)
3	Contemporaneous aggregate variables	Baseline	6.13 (1.79)	4.61 (2.53)	12.53 (2.62)	10.95 (5.18)
4	Baseline	Time-varying $MU(C)$	14.82 (3.07)	11.74 (3.60)	n.a.	n.a.
5	Time trend only	Time-varying $MU(C)$	13.83 (5.93)	13.78 (8.34)	n.a.	n.a.
6	Contemporaneous aggregate variables	Time-varying $MU(C)$	13.21 (2.96)	9.60 (3.39)	n.a.	n.a.
$\beta_{\delta,1}$						
			WLS	OLS	$1/\sigma$	
Row	First stage	Second stage	WLS	OLS	WLS	OLS
Panel B: Price equation (22)						
1	Baseline	Baseline	-1.07 (1.55)	-0.92 (1.63)	0.37 (3.50)	0.18 (4.51)
2	Time trend only	Baseline	-0.86 (0.99)	-0.30 (1.38)	-0.31 (2.18)	-0.03 (2.15)
3	Contemporaneous aggregate variables	Baseline	-0.56 (1.69)	-0.48 (1.78)	1.31 (3.88)	-0.64 (4.81)
4	Baseline	Time-varying $MU(C)$	-0.76 (1.73)	-0.57 (1.98)	n.a.	n.a.
5	Time trend only	Time-varying $MU(C)$	-0.75 (1.17)	0.13 (2.07)	n.a.	n.a.
6	Contemporaneous aggregate variables	Time-varying $MU(C)$	-0.97 (1.87)	-0.83 (2.15)	n.a.	n.a.

Notes: The baseline forecast specification (rows 1 and 4) includes a constant, trend, two lags of real GDP, real corporate profits, type-specific real investment, type-specific real relative prices, and type-specific ITC. The trend-only forecast specification (rows 2 and 5) includes only a constant and trend. The first-stage specification with contemporaneous aggregate variables is identical to the baseline forecast specification, except that date t data for GDP and corporate profits are used to forecast date t investment. The baseline structural specification (rows 1–3) uses HP-filtered consumption to measure aggregate marginal utility. The time-varying $MU(C)$ specification (rows 4–6) uses time dummies. Estimates are by ordinary least squares (OLS) or weighted least squares (WLS). Standard errors in parentheses

(a) Table 4: structural parameter estimates.

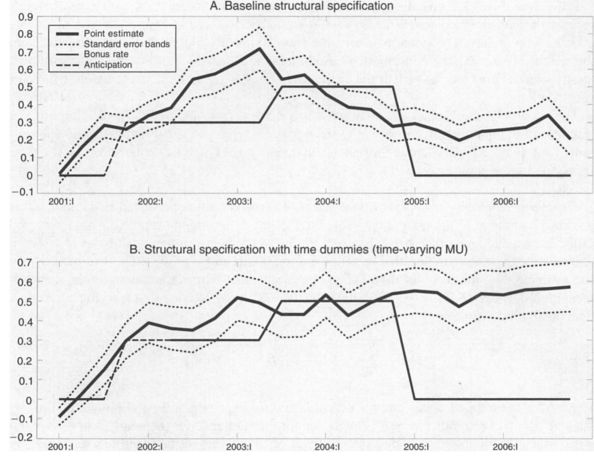


FIGURE 4. TIME-SERIES ESTIMATES OF THE BONUS DEPRECIATION RATE

Notes: The figures plot the implied time path of the best fitting bonus depreciation rate (λ_t) for the investment data in equation (23). The top panel controls for aggregate effects with Ψ_T^*C , while the bottom panel estimates a time-varying marginal utility term (a time dummy) that is then scaled across investment types by the tax term Ψ_T^* described in the text.

(b) Figure 4: time-series estimates of the implied bonus depreciation rate.

8 Short summary

- The paper studies temporary investment tax incentives using the 2002 and 2003 bonus depreciation provisions.
- The theory shows that temporary subsidies for long-lived capital should be capitalized into the shadow price of investment goods.
- Because the shadow-price response is pinned down by theory, the supply elasticity can be estimated from quantity responses.
- Bonus depreciation was more valuable for long-lived qualifying capital than for short-lived capital.
- Investment in the most subsidized capital types increased sharply.
- The estimated supply elasticity is high, approximately 6 to 14 depending on specification.
- The market price response is not clearly detectable.
- The weak price evidence may reflect internal adjustment costs or noisy price measurement.
- The strongest empirical evidence is the cross-sectional dose-response pattern across capital types.
- The time-series evidence is supportive but less sharp because of anticipation, time-to-build, Section 179 expensing, and the expiration details.

9 Conclusion

9.1 Core conclusion

House and Shapiro show that temporary investment tax incentives can generate large short-run investment responses when they apply to long-lived capital. The main mechanism is intertemporal substitution: firms move future purchases into the subsidy window. Their empirical evidence from bonus depreciation indicates a high investment supply elasticity and a strong response among the most subsidized capital categories.

9.2 Economic intuition

A temporary subsidy changes the timing of investment rather than the long-run value of installed capital. Because long-lived capital produces benefits far into the future, its shadow value is anchored by long-run fundamentals. When the after-tax purchase price temporarily falls, firms buy now instead of later. The greater the supply elasticity, the larger the observed quantity response.

9.3 Incremental contribution revisited

The paper's key contribution is to connect a simple arbitrage logic to an empirical policy experiment. It shows why price increases are not proof of inelastic supply and why quantity responses can identify the supply elasticity. The paper also demonstrates how detailed capital-type variation in the tax code can be used to estimate structural policy parameters.

9.4 Policy implications

Temporary investment incentives can be effective stimulus tools, especially when the policy is credible, short-lived, and targeted at long-lived capital. However, their effectiveness partly comes from shifting investment across time, so the long-run effect may be smaller than the short-run response. Design details - eligibility, expiration rules, and interaction with other tax provisions - matter greatly.

9.5 Limitations

The analysis relies on approximations that work best for long-lived capital and short-lived policies. The observed price data are noisy, making it difficult to distinguish internal from external adjustment costs. The time-series interpretation is complicated by anticipation, time-to-build, and Section 179 expensing. The evidence is therefore strongest for the cross-sectional treatment-intensity pattern rather than for exact timing around the policy's start and end.

9.6 Takeaway

The durable lesson is that temporary tax incentives are powerful because firms can move investment across time. Bonus depreciation increased investment in the categories it subsidized most, and the response implies a highly elastic short-run supply of investment goods.